

WEEKLY PROGRESS REPORT

Intern Name	[Your Full Name]
Week Covered	Week 2 — Data Cleaning & Exploratory Data Analysis
GitHub Repo	github.com/apekshyabasnyat22-design/hr-attribution-prediction

1,470 Rows Analyzed	0 Missing Values	16.1% Attrition Rate	15 Charts Generated
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Achievements This Week

- Profiled the full dataset (1,470 rows × 35 columns) — confirmed zero missing values across all features.
- Dynamically detected and dropped 4 non-predictive columns (EmployeeCount, Over18, StandardHours, EmployeeNumber).
- Encoded the target variable (Attrition: Yes/No → 1/0) and confirmed significant class imbalance: 83.9% stayed vs. 16.1% left.
- Ran IQR-based outlier detection across 14 numeric features; made and documented the decision to retain outliers (legitimate values, to be handled by tree-based models in Week 4).
- Completed univariate analysis (distributions) and bivariate analysis (boxplots/violin plots) for all numeric and categorical features against Attrition.
- Ran statistical T-tests confirming several feature differences between leavers and stayers are statistically significant ($p < 0.05$), not random noise.
- Generated the mandatory correlation heatmap and saved it for direct reuse in the Week 7 Streamlit Insights Tab.
- Identified top attrition drivers: OverTime (30.5% vs 10.4%), MaritalStatus (Single = 25.5%), BusinessTravel (Frequent = 24.9%), and JobRole (Sales Rep = 39.8%).
- Saved cleaned dataset (hr_attrition_cleaned.csv) and reorganized the GitHub repo into data/, notebooks/, and docs/ folders.
- Pushed all Week 2 deliverables to GitHub (notebook, cleaned data, EDA charts, EDA report).

Blockers / Challenges

- Initial environment setup required installing missingno and scipy, which were not pre-installed — resolved via pip install.
- Git push initially appeared to do nothing — root cause was files were moved locally but never staged (git add) before committing; resolved by re-running git add . → git commit → git push.
- A duplicate notebook file (with a "(2)" suffix) was accidentally created during reorganization — cleaned up before final commit to keep the repo professional.